

CMSC 471 — Introduction to Artificial Intelligence

Programming Assignment 1: A* Search and Heuristics with DOGCAT

1. Written analysis [40 points]

Submit your answers in a PDF named `pa01.pdf`.

Support your conclusions using your implementation, formal reasoning, and experimental results. A returned path may differ from the sample output while still being correct if it has the same optimal cost.

Question 1: Heuristic design and admissibility [15 points]

For each of the three cost measures—**steps**, **Scrabble**, and **frequency**—describe the heuristic implemented in your program.

For each heuristic:

1. Give a precise formula or pseudocode.
2. Explain what unavoidable remaining cost it lower-bounds.
3. Explain why it cannot overestimate the true optimal remaining cost.

Your explanation must be specific to the corresponding cost measure. Merely stating that the heuristic “is a lower bound” is insufficient.

* In all formulas c is the cumulative cost and c' is the cumulative cost after considering the current action's cost *

- Steps:

- Formula: $c + 1 = c'$
- The unavoidable remaining cost is the number of mismatched character positions. In a best case scenario in which the initial and goal do not match, the next character selected for each position would match the goal state, therefore the minimum required actions is the number of mismatched characters.
- Since the minimum number of actions is equal to the number of mismatched characters, any other solution that does not produce the goal in the number of mismatched characters will then require more actions. Therefore the heuristic cannot overestimate the true optimal remaining cost.

- Scrabble:

- Formula: $c + s = c'$ where s is the letter value of the replacement character.
- The unavoidable remaining cost are the Scrabble values of the goal letters at each mismatched position. For each mismatched character, the cost of each replacement letter is added to the cumulative cost. Therefore, at best each replacement letter is the goal state's letter at that mismatched position, requiring the cost of the goal state's letter to be added to the cumulative cost.
- The heuristic cannot overestimate the true optimal goal because it assumes that for each position mismatched, the next selected character will match the goal state, requiring only the goal state's letter value at each mismatched position to be considered.

- Frequency:

- Formula: $c + 1 + \text{rarite}(\text{goal})$
- The remaining unavoidable cost is 1 for each mismatched character position because each action can only change one character position plus the rarity value of the goal state since once it is reached, its rarity value must be considered.
- The heuristic cannot overestimate the true optimal goal because it assumes the best case where every required character correction is completed with no further actions and includes the goal's rarity value only once. Any other actions beyond the minimum would incur a greater cost, therefore the heuristic remains lower than or equal to the optimal true cost.

Question 2: Does the heuristic actually help? [10 points]

Run both **uniform-cost search** and **A*** on:

- WHY → ASK
- BEAR → DUCK

Run each problem using all three cost measures. Report:

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction	Heuristic
WHY/ASK	Steps	5	773	5	16	97.93	3
WHY/ASK	Scrabble	9	765	9	13	98.30	7
WHY/ASK	Frequency	38.8554	808	38.8554	634	21.53	7.232644
BEAR/DUCK	Steps	4	691	4	6	99.13	4
BEAR/DUCK	Scrabble	11	2981	11	6	99.80	11
BEAR/DUCK	Frequency	38.2402	1084	38.2402	302	72.14	11.22972

* The heuristic formulas from Q1 were used in their respective cells to calculate h *

Calculate the percentage reduction as:

$$100 \times \frac{\text{UCS expansions} - \text{A* expansions}}{\text{UCS expansions}}$$

1. Did UCS and A* always return the same optimal cost? If not, identify and investigate the discrepancy.
 - Yes, the optimal cost is the same in all tests.
2. For each word pair, under which cost measure did A* produce the largest percentage reduction in expanded nodes?
 - A* produced the largest percentage reduction in expanded nodes with the Scrabble cost measure for both pairs.
3. For each word pair, under which cost measure was the heuristic least informative?
 - For both pairs, the frequency cost measure had the least informative heuristic.
4. Explain the differences using the definitions of your heuristics, not runtime alone.
 - Steps is highly informative because each action costs the same, 1, thus the absolute minimum cost required is the number of mismatched characters.
 - Scrabble is also highly informative because the similar to steps, it can easily account for the absolute minimum cost required to achieve goal by considering the minimum of actions required is tied to the number of mismatched characters and their replacement values are known beforehand.
 - Frequency on the other hand is the least informative by a large margin due to the very nature that only the minimum number of actions is known. The effect of the rarity of each intermediary word reached before goal can cause much bigger impacts than just accounting for the goal's rarity value.

Different optimal paths are acceptable when their total costs are equal.

Question 3: Reversibility and direction-dependent costs [5 points]

DOGCAT actions are reversible: if one legal word can be changed into another, the reverse change is also legal.

For each cost measure—**steps**, **Scrabble**, and **frequency**—determine whether the optimal cost from W_1 to W_2 must always equal the optimal cost from W_2 to W_1 .

For each cost measure:

1. State **yes** or **no**.
2. Justify your conclusion.
3. If the answer is no, give a concrete counterexample using words from the provided dictionary.

Clearly distinguish between the reversibility of an action and the symmetry of its cost.

- Steps: Yes, because the number of mismatched positions and the cost of each action remains the same, the solution path produced would therefore cost the same whether you start from W_1 or W_2 .
- Scrabble: No, because each letter may not have the same letter-value. For example:

$$\text{cost}(\text{pay}, \text{way}) = 4$$

$$\text{cost}(\text{way}, \text{pay}) = 3$$

As we can see, since the replacement value of c and d differ, the optimal cost will differ.

- Frequency: No, because similar to the reasoning for steps, the rarity value of the initial and goal states may also differ. Therefore in the best case scenario in which only one action is required to go from initial to goal, the total optimal cost may differ. For example:

$$\text{cost}(\text{get}, \text{got}) = 1 + \text{rarity}(\text{got}) = 1 + 2.710171 = 3.710171$$

$$\text{cost}(\text{got}, \text{get}) = 1 + \text{rarity}(\text{get}) = 1 + 1.8609667 = 2.8609667$$

Question 4: LLM heuristic audit [10 points]

Give a generative AI system the following prompt:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary. The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

A simple way to make h_0 rarity-aware is to exploit a cost that every solution must pay: the cost of entering the goal word.

For a state n , let

$$d(n, g) = \#\{i: n_i \neq g_i\}$$

be the Hamming distance from n to the goal g . Define

$$h_R(n) = d(n, g) + r(g)$$

for $n \neq g$, where $r(g)$ is the dictionary's rarity value for the goal. At the goal itself, use

$$h_R(g) = 0.$$

1. OpenAI GPT-5.6 Luna
3. The LLM reached the same conclusion as my response found in Q1. If the initial state is not the goal state, then the minimum number of actions required is equal to the number of mismatched characters. Then since the initial state did not match the goal state, at minimum the rarity value of the goal state must be considered. Therefore, the proposed heuristic by the LLM is admissible.
4. The proposal is valid.

5.

Word pair	Heuristic	Solution Cost	Nodes Expanded
WHY/ASK	h_0	38.85541	755
WHY/ASK	h	38.85541	634
BEAR/DUCK	h_0	38.24023	720
BEAR/DUCK	h	38.24023	302

As seen from testing, there is evidence that the proposed heuristic reduced the number of expanded nodes while obtaining the same optimal cost solution.